Research Paper Summary Submission

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Predictive Models with XAI: A Comparative Study of Enhancing Airline Customer Satisfaction

Authors: Cloë C. E. van Geest, Yong Wan Yit, Zaur T. Gouliev, and Keith Quille | Year: 2023
Source/Conference/Journal/Publisher: Human Centered Al Education and Practice Conference 2023 (HCAlep '23), ACM, New York, NY, USA

The research paper addresses the critical need for predicting airline customer satisfaction in the competitive airline industry. The study's primary objective is to use explainable AI (XAI) to predict customer satisfaction based on various parameters and to identify areas where airlines can improve their services to generate more satisfied customers. The authors acknowledge that predicting satisfaction is a complex task due to the numerous factors that influence it, such as passenger demographics, flight details, and in-flight services.

The researchers used an **Airlines Customer Satisfaction dataset** from IIT Roorkee, which contains 129,880 rows and 24 columns, including the target variable "satisfaction". To tackle the problem, they employed two distinct approaches: a **Blackbox approach** using a deep neural network and a **Glassbox approach** using a decision tree. The Blackbox model was built with TensorFlow and Keras, featuring several hidden layers with ReLU activation functions and an output layer with a sigmoid activation function for binary classification. For the Blackbox model, they incorporated XAI techniques, specifically **LIME** and **SHAP**, to gain insights into its predictions and feature importance. The Glassbox model, being interpretable on its own, didn't require XAI.

The results showed that the Blackbox model achieved an overall accuracy of **92%**, while the Glassbox model reached a higher accuracy of **94%**. The Glassbox model also had a higher recall and a comparable F1-score, making it more balanced and effective across performance metrics. Both LIME and SHAP analyses on the Blackbox model consistently identified **"Seat Comfort"** and **"Inflight Entertainment"** as the most significant predictors of customer satisfaction. Surprisingly, the XAI analyses also highlighted "Gender" as a significant factor. The study concludes that for this dataset, which is of low complexity, the Glassbox approach is recommended due to its similar accuracy and inherent explainability, which eliminates the need for additional XAI techniques.

A limitation of the study is that the predictive model is static and based on historical data, which may not account for real-time fluctuations in the airline industry. The authors suggest that future work should focus on creating dynamic models capable of adapting to these changes. The authors also suggest further research to understand how gender influences customer satisfaction.

FairAD-XAI: Evaluation Framework for Explainable Al Methods in Alzheimer's Disease Detection with Fairness-in-the-loop

Authors: Quoc-Toan Nguyen, Linh Le, Xuan-The Tran, Thomas Do, and Chin-Teng Lin | Year: 2024 | Companion of the 2024 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp Companion '24), ACM, New York, NY, USA

The paper "FairAD-XAI" introduces a framework to address the challenging evaluation of Explainable Artificial Intelligence (XAI) methods, particularly for Alzheimer's Disease (AD) detection using low-cost or wearable devices. The authors highlight a significant gap in current XAI evaluations, which often lack input from medical professionals and may lead to biased decisions that unfairly impact minority groups. The core of their proposed solution is a validation framework called 'FairAD-XAI,' which provides a comprehensive assessment through twelve properties of explanations, structured into a detailed Likert questionnaire.

The methodology of the **FairAD-XAI framework** involves two key components: a **Likert questionnaire** and **fairness-in-the-loop evaluation metrics**. The questionnaire, based on "co-twelve" explanation quality properties, is grouped into three dimensions: Content, Presentation, and User. These questions are designed to assess how accurately, concisely, and appropriately an explanation is for a given user. For instance, the **Content** dimension evaluates correctness, completeness, consistency, continuity, contrastivity, and covariate complexity.

To ensure fairness, the framework integrates two fairness metrics: **Disparate Impact (DI_{XAI})** and **Demographic Parity (DP_{XAI})**. These metrics measure whether different demographic groups receive equitable and unbiased explanations. For example, DI_{XAI} is the ratio of average scores between different demographic groups, while DP_{XAI} calculates the absolute difference in average scores. The framework provides a clear implementation algorithm for how these metrics are calculated and used to label an XAI method as "fair" or "unfair" based on a set threshold.

As a case study, the authors developed a binary AI model for AD detection using the Alzheimer's Disease Neuroimaging Initiative (ADNI) dataset, incorporating five cognitive and functional tests. They compared five machine learning methods (DT, RF, SVM, KNN, XGBoost) and found that **Random Forest (RF)** was the best-performing model. They then used the RF model with the OmniXAI library, which utilizes **LIME** for explanations, to demonstrate how the FairAD-XAI framework can be applied to evaluate the fairness and quality of the explanations. The example demonstrates the framework's use with sensitive attributes like sex, race, and marital status. The conclusion is that FairAD-XAI is a vital and versatile tool for evaluating XAI methods, not only for AD detection but also for other domains, as it emphasizes explanation quality and fairness across different user demographics.